

# Functional System Design (FSD) Documentation

Implementation-oriented documentation for Incident Life Cycle Automation

| DATE            | STUDENT NAME | PROJECT                        |
|-----------------|--------------|--------------------------------|
| 07 October 2026 | Gangadevi.R  | Incident Life Cycle Automation |

## 1. Introduction

Project Title: Incident Life Cycle Automation

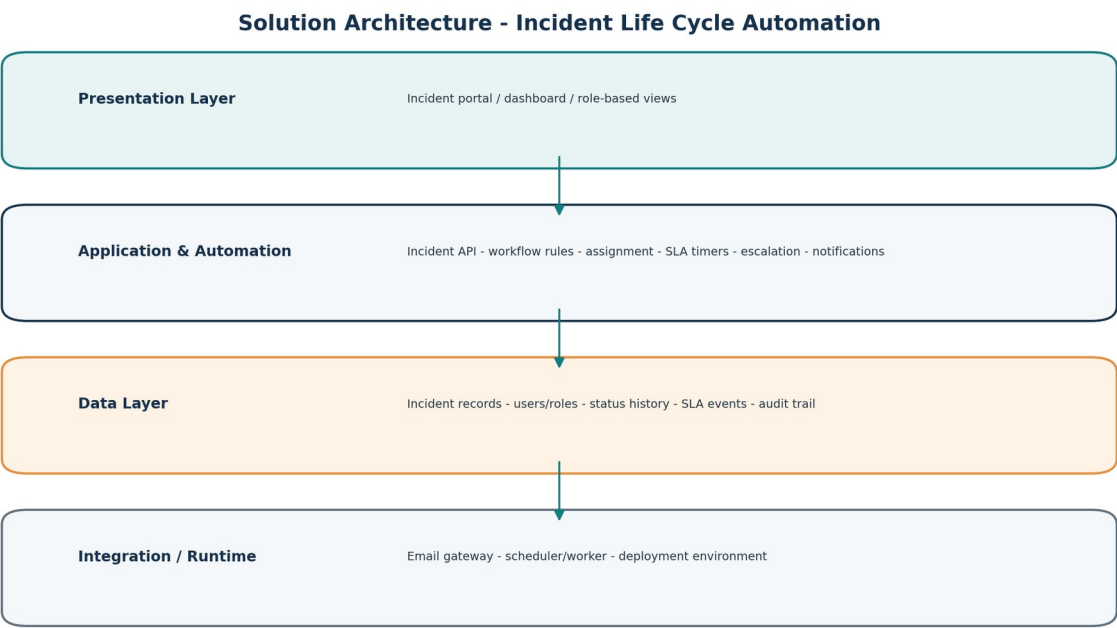
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## 2. Project Overview

Purpose: automate the flow of operational incidents from intake to closure while making ownership, priority, SLA state, notifications and audit history visible and consistent.

- Incident intake and unique identification
- Rule-based triage and assignment
- SLA monitoring and escalation
- Lifecycle status tracking and work notes
- Resolution and closure
- Search, dashboards and audit trail

### 3. Architecture



### 4. Data model (logical)

| Entity         | Key data                                                                                                | Relationship                                                        |
|----------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Incident       | Incident ID, title, description, source, category, impact, urgency, priority, status, owner, timestamps | Central record linked to users, history, SLA events and resolution. |
| User / Role    | User ID, name, role, access scope                                                                       | Controls who can view or modify incidents.                          |
| Status History | Incident ID, old status, new status, actor, timestamp, note                                             | One-to-many with Incident.                                          |
| SLA Event      | Incident ID, due time, event type, triggered time, state                                                | Tracks SLA and escalation lifecycle.                                |
| Audit Entry    | Actor, action, entity, timestamp, context                                                               | Traceability for key changes.                                       |

### 5. API / Service responsibilities

| Service          | Purpose                                   |
|------------------|-------------------------------------------|
| Incident Service | Create, read, update and close incidents. |
| Triage Service   | Apply classification and priority rules.  |

| Service              | Purpose                                        |
|----------------------|------------------------------------------------|
| Assignment Service   | Resolve queue/owner and manage reassignment.   |
| SLA Worker           | Check due times and emit escalation events.    |
| Notification Service | Send assignment, status and SLA notifications. |
| Audit Service        | Persist lifecycle and administrative history.  |

## 6. Testing

Functional testing should cover positive and negative flows; UAT validates user expectations; performance testing measures response and scheduled-job behavior using actual implementation results.

## 7. Known limitations

- Initial scope prioritizes core incident lifecycle automation and excludes unrelated features from the current build.
- Actual performance figures depend on the final deployment environment and dataset size.

## 8. Future enhancements

- Multiple SLA policies by service and priority
- Advanced analytics and trend reporting
- Integrations with enterprise ticketing / messaging tools